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Pontoon/tritoon watercraft with composite chassis

Abstract

A watercraft includes: a watercraft body; and a pontoon frame system coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; and a chassis coupled with and at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end.

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Background/Summary

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to boats using two or more pontoons, and, more particularly, to the chassis of such boats.
2. Description of the Related Art
(2) Certain boats (which can be referred to herein as watercraft) may include two or more pontoons as floatation devices. Boats that use two pontoons are herein referred to as pontoon boats, and boats that use three pontoons are herein referred to as tritoon boats. Pontoons are tubes (which can be referred to as floatation tubes) or hulls that are airtight and hollow so that they are able to float, and they are made of known materials.
(3) Existing pontoon/tritoon watercraft experience problems with their chassis design. More specifically, existing pontoon/tritoon watercraft chassis designs require at least 100 man-hours to construct, suffer from poor tolerances in the construction, and experience excessive twisting of the pontoon frame system under high speed and/or rough water conditions. Such twisting results in weakening of the connective fasteners attached to the flotation tubes and the chassis, as well as of the connections between the chassis, the flooring, the furniture, the helm, the railings, the trim, the motor mounts, and other structures mounted above the chassis. Additionally, such twisting creates

an unpleasant ride for the passengers, thus diminishing the ride experience.
(4) Such chassis designs have a plurality of transverse beams coupled with each of the pontoons. Each beam extends transversely across, and thus perpendicular to, the pontoons. The pontoons extend longitudinally running fore to aft. Further, a forward-most transverse beam is attached to the forward end of the pontoons, a rearward-most transverse beam is attached to the rearward end of the pontoons, and a plurality of transverse beams are spaced apart between these forward-most and rearward-most transverse beams.
(5) What is needed in the art is an improved, cost-effective way to mitigate the aforementioned problems associated with pontoon/tritoon watercraft.
SUMMARY OF THE INVENTION
(6) The present invention provides a pontoon/tritoon chassis including a composite structure with various attachment features.
(7) The invention in one form is directed to a pontoon frame system of a watercraft including a watercraft body, the pontoon frame system being coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; and a chassis configured for being coupled with and for at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end.
(8) The invention in another form is directed to a watercraft including: a watercraft body; and a pontoon frame system coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; and a chassis coupled with and at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end.
(9) The invention in yet another form is directed to a method of using a watercraft, the method including the steps of: providing that the watercraft includes: a watercraft body; and a pontoon frame system coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; and a chassis coupled with and at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end; and floating the watercraft.
(10) An advantage of the present invention is that it provides for making a pontoon/tritoon watercraft chassis with considerably lower cost and ease of build.
(11) An advantage of the present invention is that it provides a pontoon/tritoon watercraft chassis that is easier to maintain.
(12) Yet another advantage of the present invention is that it provides a pontoon/tritoon watercraft chassis that is more rigid and stable under rough water and/or high-speed operation. The chassis provides stability to an entire platform, eliminating twisting and flexing associated with known pontoon/tritoon watercraft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention will be better understood by reference

to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

(2) FIG. **1** is a top, perspective view of an exemplary embodiment of a watercraft, the watercraft including a pontoon frame system including pontoons and a chassis with a composite body, in accordance with an exemplary embodiment of the present invention;

(3) FIG. **2** is a bottom, perspective view of the chassis with the composite body of FIG. **1**, in accordance with an exemplary embodiment of the present invention;

(4) FIG. **3** is a top, perspective view of another exemplary embodiment of the chassis of the watercraft, in accordance with an exemplary embodiment of the present invention;

(5) FIG. **4** is a bottom, perspective view of the chassis of FIG. **3**;

(6) FIG. **5** is a top, perspective view of yet another exemplary embodiment of the chassis of the watercraft, in accordance with an exemplary embodiment of the present invention;

(7) FIG. **6** is a bottom, perspective view of the chassis of FIG. **5**; and

(8) FIG. **7** is a flow diagram showing a method of using a watercraft, in accordance with an exemplary embodiment of the present invention.

(9) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE INVENTION

(10) The terms “forward”, “rearward”, “left” and “right”, when used herein in connection with the watercraft and/or components thereof, are usually determined with reference to the direction of forward operative travel of the watercraft, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the watercraft and are equally not to be construed as limiting. The terms “downstream” and “upstream” are analogous to “rearward” and “forward,” respectively.

(11) Referring now to the drawings, and more particularly to FIG. **1**, there is shown a watercraft **100** (which can be referred to herein as a boat **100**) which generally includes a watercraft body **102** and a pontoon frame system **104** coupled with the watercraft body **102** (such as by way of fasteners **324**, discussed below). The watercraft body **102** is shown schematically and can include the parts of the watercraft mounted primarily atop the pontoon frame system **304**; these parts can include, by way of example and not limitation, a floor, decking, a platform, railings, furniture, a helm, and any other structures that are mounted on top of the floor. Though shown relatively small in FIG. **1**, watercraft body **102** can span at least the entire longitudinal and transverse extent of chassis **108**.

(12) Pontoon frame system **104** incudes a plurality of pontoons **106** and a chassis **108**. Pontoons **106** are spaced apart from one another in a transverse direction and extend in a longitudinal direction of watercraft **100**. As is known, watercraft **100** with pontoons **106** typically includes two pontoons **106** (a pontoon boat) or three pontoons **106** (a tritoon boat). Though the figures show watercraft **100** of the present invention formed as a tritoon boat, it can be readily appreciated that the watercraft of the present invention can be formed as a pontoon boat in the figures by removing the middle pontoon (as well as features of watercraft **100** associated with the middle pontoon **106**). Pontoons **106** include a forward end **110**, a rearward end **112**, a first outboard pontoon **114**, and a second outboard pontoon **116**. Regarding a tritoon boat (as shown), the outboard pontoons **114**, **116** are the transverse side pontoons **106**. It can be appreciated that if the watercraft has only two pontoons **106** (a pontoon boat), then the two pontoons **106** can still be deemed to be first and second outboard pontoons, respectively. Pontoons **106** include risers **120** attached thereto.

(13) Chassis **108** of pontoon frame system **104** is coupled with and at least partially supports watercraft body **102**. Chassis **108** can be coupled with watercraft body **102** in any suitable manner, though one way is shown and discussed below with respect to fasteners **324** in conjunction with FIGS. **3-6**. Further, chassis **108** is supported by and couples together pontoons **106**. Chassis **108** includes a composite body **118** (which can include a matrix material (such as a resin) with fibrous

material embedded therein and can be referred to as a unibody) coupled with each pontoon **106** at forward end **110** and rearward end **112** of pontoons **106**. Composite body **118** can have any suitable shape, such as a slab (shown) or the like. Composite body **118** can be made of any suitable material and in any suitable way. Such materials include, but are not limited to, fiberglass, carbon fiber, polymers, plastics, other composites, castable metals, and/or the like. By way of example and not limitation, one option for the material is a carbon fiber-based product, made by Vectorply of Phenix City, Alabama, the product being marketed as a VECTORULTRA™ with a product code of C-QX **2300**. Composite body can be molded, formed, and/or cast, for example.

(14) To couple together composite body **118** with pontoons **106**, a plurality of risers **120** attached to pontoons **106** can be slidably received in corresponding slots **122** formed in composite body **118**. Risers **120** can be made of any suitable material, such as a metal (such as aluminum, or steel), a polymer, carbon fiber, and/or fiberglass, and can be made in any suitable manner. Risers **120** are coupled with pontoons **106** in any suitable manner, such as by way of brackets, fasteners, and/or welding, and any devices making these connections can be made of any suitable material, such as aluminum, steel, carbon fiber, fiberglass, and/or the like. A single riser **120** can be attached to a respective platoon **106** and run substantially the length of the respective pontoon **106** from forward end **110** to rearward end **112** of the pontoon **106**. Alternatively, rather than a single, long riser **120**, a plurality of risers **120**—shorter in their longitudinal extent—can be attached to and spaced longitudinally apart from one another along the length of the respective pontoon **106** from forward end **110** to rearward end **112** of the pontoon **106**. Each riser **120** can be formed with a right-angle and thus have substantially an L-shape. Further, composite body **118** can include slots **122** which matingly receive respective risers **120**. Slots **122**, as shown, can thus have a right-angle and thus have substantially an L-shape. These slots **122** can be formed in composite body **118** when composite body **118** is, for example, molded, or can be cut or machined into composite body **118** subsequent to molding of composite body **118**. During assembly of pontoon frame system **104**, risers **120** can be slidably received by slots **122**, either before or after risers **120** are attached to pontoons **106**.

(15) Referring now to FIG. **2**, there is shown a bottom, perspective view of chassis **108** of watercraft **100**, with portions broken away. Chassis **108** is thus shown by itself, without risers **120** connected thereto. Chassis **108** is shown to include composite body **118** with three slots **122** corresponding to three pontoons **106**. Each slot **122**, as shown, can extend the longitudinal extent of composite body **118**. Each slot **122**, with its right-angle formation, is configured to matingly receive risers **120**. During assembly, risers **120** can be inserted into a respective slot **122** at either end of composite body **118**. Securing risers **120** within slots **122** can occur in any suitable manner, such as by way of a press fit, an interference fit, using an adhesive, or the like.

(16) Referring now to FIG. **3**, there is shown a top, perspective view of another embodiment of the chassis of watercraft **100**, according to the present invention. Certain prior reference numbers with respect to the chassis are increased by a multiple of 100 and are substantially similar to the structures and function of the embodiments shown in FIGS. **1-2**, unless otherwise shown and/or described. Thus, the chassis and composite body are now labeled **308** and **318**, respectively, in FIG. **2**. One substantial difference between chassis **108** and chassis **308** is that composite body **318** includes a plurality of fasteners **324** attached thereto, which can be made of any suitable material, such as aluminum, steel, carbon fiber, fiberglass, and/or the like. More specifically fasteners **324** are embedded in the composite material of composite body **318**; such fasteners **324** can be referred to as captivated machine screw nuts **324**, and in FIG. **3** fasteners **324** are embedded in a top portion of composite body **318**. Alternatively, rather than being screw nuts, fasteners **324** can be screws, bolts, or the like. Fasteners **324** serve to mount all or portions of watercraft body **102**. Thus, by way of example, fasteners **324** can serve to mount the floor, helm, railings, furniture, and/or any other structures that are required to be mounted on top of the floor.

(17) Another substantial difference is that composite body **318** includes a high-voltage battery pack

326 embedded therein. Pack **326** is thus captured within composite body **318**, which is thick enough to embed such a pack **326**. In this way, pack **326**, being under the floor (part of watercraft body **102**), is isolated from passengers above the floor. Access can be had to pack **326** in any suitable manner. Further, though pack **326** is shown in FIG. **3** relatively small and placed between fasteners **324**, this is done only for illustrative purposes; it can be appreciated that pack **326** can be larger than what is shown in FIG. **3** (as large as needed) and can be positioned under and/or above fasteners **324**. Pack **326** can be formed of any suitable material, and may include a substantially rigid shell or enclosure. Pack **326** can be used, for example, as part of an all-electric boat design (a boat powered by electricity). Pack **326** can include therein one or more batteries as is suitable for the intended use. Further, any or all high voltage and high current cables can be at least partially contained or encapsulated within pack **326** and/or routed underneath the flooring of watercraft body **102** as well, eliminating the risk of accidental electrocution. Any such cables may route to an inverter and/or an outboard motor. Further, any such cables, if not in pack **326**, can be routed in composite body **318** and/or within recesses **528** (below). It can be appreciated that pack **326** can be included in any of the embodiments of the present invention shown herein.
(18) Referring now to FIG. **4**, there is shown a bottom, perspective view of chassis **308** of FIG. **3**. Chassis **308** is shown to include composite body **318** with three slots **122** corresponding to three pontoons **106**. Each slot **122**, as shown, can extend the longitudinal extent of composite body **318**. Further, additional fasteners **324** are embedded on a bottom portion of composite body **318**, similar to how fasteners **324** are embedded on the top portion of composite body **318**, though the positions of the fasteners **324** on the bottom portion of composite body **318** can be in locations which do not correspond to the positions of the fasteners **324** on the top portion, as shown when comparing FIGS. **3** and **4**. These fasteners **324** on the bottom portion are configured to mount various structures, for example, trim pieces, motor supports, and any other suspended structures that require attachment to the bottom of composite body **318**.
(19) Referring now to FIG. **5**, there is shown a top, perspective view of another embodiment of the chassis of watercraft **100**, according to the present invention. Certain prior reference numbers with respect to the chassis are increased by a multiple of 100 and are substantially similar to the structures and function of the embodiments shown in FIGS. **1-4**, unless otherwise shown and/or described. Thus, the chassis and composite body are now labeled **508** and **518**, respectively, in FIG. **5**. Composite body **518** includes slots **122** and fasteners **324** embedded therein. One substantial difference between chassis **308** and chassis **508** is that composite body **518** includes at least one recess **528** configured for receiving and, as is necessary, routing a device **530** therein (device **530** is shown schematically). FIG. **5** shows three such recesses formed in the top portion of composite body **518**. Device **530** can be any suitable device, including, but not limited to cooling lines, electrical conduits, lighting cables or tubing, stereo cables or tubing, and/or any other cabling or tubing that requires routing. Recesses **528** are thus configured for receiving and routing therein any such device **530**.
(20) Referring now to FIG. **6**, there is shown a bottom, perspective view of chassis **508** of FIG. **5**. Chassis **508** is shown to include composite body **518** with three slots **122** corresponding to three pontoons **106**. Each slot **122**, as shown, can extend the longitudinal extent of composite body **518**. Further, additional fasteners **324** are embedded on the bottom portion of composite body **518**. The bottom portion of composite body **518** also includes, like the top portion, recesses **528**, which can be located in different positions relative to recesses **528** in the top portion of composite body **518**. Recesses **528** on the bottom portion are configured for receiving and, as is necessary, routing therein device **530**, as described above.
(21) In sum, the present invention has been made in view of the complexity, material cost (typically) aluminum, and man-hours required to measure, cut, drill, and mount all of the required hardware to attach pontoons **106** (the floatation tubes) on the bottom of chassis **508**, including splash pans, motor mounts, conduit and piping and other structures under the flooring, as well as

the furniture, railings, trim pieces, motor mounts, helm and all other fixtures mounted on top of the floor (summarized as watercraft body **102** and/or device **530**, depending upon whether fasteners **324** or recesses **528** are used).
(22) In use, during manufacture, chassis **108**, **308**, **508** can be formed including a formed, molded, or cast (or other suitable way of manufacturing) structure, namely, composite body **118**, **318**, **518**. Composite body **118**, **318**, **518** can be molded, for example, to include slots **122** and recesses **528** therein, and to embed fasteners **324**. Any suitable manufacturing process can be used, such as any suitable forming process, molding process, or casting process, for example. Risers **120** can be slid into slots **122**, either before or after attaching risers **120** to pontoons **106**. Further, any additional structures associated with watercraft body **102** and/or device **530** can be connected to fasteners **324** and stored or routed in recesses **528**, respectively. Devices **530** stored or routed in recesses **528** can be attached to composite body **518** in any suitable manner. Upon assembling watercraft **100**, watercraft **100** can be floated or otherwise used on a body of water.
(23) Referring now to FIG. **7**, there is shown a flow diagram showing a method **740** of using watercraft **100**. Method **740** includes the steps of: providing **742** that the watercraft **100** includes: a watercraft body **102**; and a pontoon frame system **104** coupled with the watercraft body **102**, the pontoon frame system **104** including: a plurality of pontoons **106** spaced apart from one another and including a forward end **110**, a rearward end **112**, a first outboard pontoon **114**, and a second outboard pontoon **116**; and a chassis **108**, **308**, **508** coupled with and at least partially supporting the watercraft body **102**, the chassis **108**, **308**, **508** supported by and coupling together the plurality of pontoons **106**, the chassis **108**, **308**, **508** including a composite body **118**, **318**, **518** coupled with the first outboard pontoon **114** and the second outboard pontoon **116** at the forward end **110** and the rearward end **112**; and floating **744** the watercraft **100**. The plurality of pontoons **106** can include a plurality of risers **120** attached thereto, the composite body **118**, **318**, **518** including a plurality of slots **122** which slidably receive the plurality of risers **120**. The composite body **318**, **518** can include a plurality of fasteners **324** embedded therein. The composite body **518** can include at least one recess **528** configured for receiving a device **530** therein. The composite body **118**, **318**, **518** can include fiberglass, carbon fiber, and/or a polymer. The composite body **318** can include a high-voltage battery pack **326** therein.
(24) While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

Claims

1. A pontoon frame system of a watercraft including a watercraft body, the pontoon frame system being coupled with the watercraft body, the pontoon frame system comprising: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; a chassis configured for being coupled with and for at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end; and a plurality of risers attached to the plurality of pontoons, the composite body including a plurality of slots which slidably receive the plurality of risers.
2. The pontoon frame system of claim 1, further including a plurality of fasteners embedded in the composite body.
3. The pontoon frame system of claim 2, wherein the composite body includes at least one recess

configured for receiving a device therein.
4. The pontoon frame system of claim 1, wherein the composite body includes at least one of fiberglass, carbon fiber, and a polymer.
5. The pontoon frame system of claim 1, further including a high-voltage battery pack embedded in the composite body.
6. A watercraft, comprising: a watercraft body; and a pontoon frame system coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; a chassis coupled with and at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end; and a plurality of risers attached to the plurality of pontoons, the composite body including a plurality of slots which slidably receive the plurality of risers.
7. The watercraft of claim 6, wherein the pontoon frame system further includes a plurality of fasteners embedded in the composite body.
8. The watercraft of claim 7, wherein the composite body includes at least one recess configured for receiving a device therein.
9. The watercraft of claim 6, wherein the composite body includes at least one of fiberglass, carbon fiber, and a polymer.
10. The watercraft of claim 6, further including a high-voltage battery pack embedded in the composite body.
11. A method of using a watercraft, the method comprising the steps of: providing that the watercraft includes: a watercraft body; and a pontoon frame system coupled with the watercraft body, the pontoon frame system including: a plurality of pontoons spaced apart from one another and including a forward end, a rearward end, a first outboard pontoon, and a second outboard pontoon; a chassis coupled with and at least partially supporting the watercraft body, the chassis supported by and coupling together the plurality of pontoons, the chassis including a composite body coupled with the first outboard pontoon and the second outboard pontoon at the forward end and the rearward end; and a plurality of risers attached to the plurality of pontoons, the composite body including a plurality of slots which slidably receive the plurality of risers; and floating the watercraft.
12. The watercraft of claim 11, wherein the pontoon frame system further includes a plurality of fasteners embedded in the composite body.
13. The watercraft of claim 12, wherein the composite body includes at least one recess configured for receiving a device therein.
14. The watercraft of claim 11, wherein the composite body includes at least one of fiberglass, carbon fiber, and a polymer.
15. The watercraft of claim 11, wherein the pontoon frame system further includes a high-voltage battery pack embedded in the composite body.